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# Satellite-derived spread dynamics of three-dimensional fire events in Sudan

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Fire is a recurrent, strongly seasonal disturbance across Sudan's savannas, yet aggregate burned-area statistics cannot resolve long-term changes in the expansion of fire events. Here, we identified individual fire events across Sudan from August 2002 to July 2025 by grouping burned MODIS MCD64A1 pixels according to spatial and temporal connectivity criteria. For each event, we calculated the mean areal spread rate as the total burned area divided by event duration ($km^2$ $day^{-1}$). Sensitivity tests using a Global Fire Atlas reference event indicated that a 3-pixel/3-day connectivity setting balanced fragmentation and excessive merging. Retaining events lasting $\geq$3 days and burning $\geq$20 $km^2$ yielded 13,290 events, concentrated in the southern part of Sudan. Across 23 fire seasons, total burned area declined significantly, whereas mean event area showed no significant trend. Median and 10th-percentile areal spread rates each decreased by approximately 9.4% $decade^{-1}$ ($p < 0.001$), while the 90th-percentile trend was weaker and non-significant ($p = 0.091$). *K*-means clustering revealed two broad propagation types: a more frequent lower-expansion type and a rarer higher-expansion type with greater areal spread rates and displacement. The decline was concentrated in lower-expansion dry-season events, with no significant trends among higher-expansion events or wet-season events. These findings show that fire-event expansion in Sudan has declined selectively rather than uniformly, revealing changes that aggregate burned-area trends alone cannot capture.

KEYWORDS
**areal spread rate, fire-event detection, savanna fire, Sudan, three-dimensional connected-component algorithm**

## 1 Introduction

Fire is a fundamental ecological process that shapes ecosystem structure, biome distributions, global biogeochemical cycles, and human societies (Pausas et al., 2025). Fire regimes are commonly characterized using aggregate measures such as fire frequency, total burned area, and seasonality, together with event-level attributes such as fire size, duration, and areal spread rate (Mahood et al., 2025). Aggregate measures quantify the overall magnitude and timing of fire activity but do not reveal how burned area is organized among individual fire events. Regions with similar total burned area can differ substantially in the number, size, duration, and areal spread rates of the fires. Analyses based only on burned pixels or regional totals can therefore obscure changes in how individual fires develop through space and time.

Satellite burned-area products provide spatially explicit, multi-decadal records that enable event-level analysis over large regions (Giglio et al., 2018). The MODIS MCD64A1 product (Collection 6.1) maps burned areas at a nominal spatial resolution of 500 m and assigns an estimated burn date to each detected burned pixel, making it possible to reconstruct the approximate spatiotemporal development of individual fires (Giglio et al., 2018). To derive event-level metrics, however, these pixel-level detections must first be grouped into discrete fire events. A three-dimensional connected-component algorithm addresses this challenge by linking burned pixels according to specified spatial and temporal connectivity thresholds (Balch et al., 2020), while the Global Fire Atlas provides an alternative global event-tracking product derived from the MODIS burned-area record (Andela et al., 2019). The resulting event population is sensitive to the selected connectivity parameters. Restrictive thresholds can fragment a single fire into multiple events, whereas permissive thresholds can merge distinct fires that should remain separate (Balch et al., 2020). This trade-off is especially consequential in frequently burned savannas, where closely spaced ignitions and nearly continuous burned patches complicate the separation of adjacent events (Andela et al., 2019).

Once individual fire events have been delineated, their propagation can be characterized by event duration, burned area, areal spread rate, and displacement (Andela et al., 2019; Mahood et al., 2025). These metrics describe complementary aspects of fire expansion and movement, and their joint variation can distinguish recurrent propagation behaviors within heterogeneous event populations. Global analyses have shown that large fires contribute disproportionately to total burned area and display propagation characteristics distinct from those of smaller events (Andela et al., 2019). Multivariate classification can further identify broad propagation types from combinations of spread-rate and displacement attributes (Archibald et al., 2013). Nevertheless, most event-based studies have emphasized spatial patterns or climatological characteristics, while long-term changes in event propagation remain less well characterized, particularly at national and regional scales (Chen et al., 2022). Such changes may differ across the spread-rate distribution: trends among slowly expanding or typical events need not match those among rapidly expanding events, and changes in the mean may conceal this divergence. Evidence of contrasting changes in mean and upper-tail fire behavior in forested systems (Juang et al., 2022; Cunningham et al., 2024) raises the question of whether similar distributional changes occur in savannas, where fuel structure, ignition patterns, and climatic controls differ substantially.

African savannas account for the majority of global burned area and are thus central to global fire-climate-vegetation interactions (Jones et al., 2022). Fire activity across the continent is strongly seasonal and closely tied to the dry season, when senesced herbaceous fuels become flammable and fuel connectivity increases (Archibald et al., 2009; Eames et al., 2023). Total burned area has declined significantly across many African savanna regions over recent decades, with cropland expansion, agricultural intensification, landscape fragmentation, changes in dry-season length, and interacting climatic and anthropogenic controls contributing to the observed trend (Andela and van der Werf, 2014; Andela et al., 2017; Jones et al., 2022; Mohammed et al., 2026). Existing assessments have focused primarily on burned area, fire frequency, seasonality, and carbon emissions (Andela and van der Werf, 2014; Andela et al., 2017; Ramo et al., 2021). Yet a decline in total burned area can result from fewer fire events, smaller events, slower areal expansion, or different combinations of these processes. Resolving these alternatives requires a long-term event-based assessment that separates changes in total fire activity from changes in the behavior of individual fires.

Sudan provides a useful setting for such an assessment because it spans a north–south hydroclimatic and vegetation gradient. Conditions range from the hyper-arid southern margin of the Sahara, with annual precipitation of approximately 100 mm, to wetter savanna and woodland environments in the south, where annual precipitation reaches approximately 700–1,200 mm (Arliah et al., 2021; Red Cross Red Crescent Climate Centre, 2024; Mohieldeen and Hassan, 2025). This gradient produces large spatial differences in fuel availability, fuel continuity, and fire-season length. Fire activity is concentrated in the southern part of the country and is highly seasonal. For example, a MODIS-based study in southern Gedaref State found that fire occurred mainly from mid-September to late April, with burned area peaking in October and November (Modawi et al., 2015). However, previous research in Sudan has largely examined seasonality, fire frequency, burned area, and carbon emissions (Modawi et al., 2015; Ramo et al., 2021). To our knowledge, no national-scale study has jointly assessed the propagation types and multi-decadal spread-rate trends of individual fire events. It therefore remains unknown whether event-level fire expansion has changed alongside the broader decline in African burned area, and whether any changes differ across the spread-rate distribution, propagation types, and seasons.

Here, we investigate whether long-term changes in aggregate burned area are accompanied by uniform changes in event-level areal expansion, or whether these changes differ across the spread-rate distribution, propagation types, and seasons. We address this question by applying a spatiotemporal connectivity approach to the MODIS MCD64A1 burned-area record for Sudan from August 2002 to July 2025. We first assess the sensitivity of event delineation to spatial and temporal connectivity settings using Global Fire Atlas reference events. We then characterize fire-event patterns, identify broad descriptive propagation types using K-means clustering, and examine long-term changes in the 10th, 50th, and 90th percentiles of mean areal spread rate, including contrasts between propagation types and between dry- and wet-season events. This analysis distinguishes changes in total burned area from changes in how rapidly individual fire events accumulate burned area.

# 2 Materials and methods

## 2.1 Study area and datasets

Sudan is located in northeastern Africa, extending approximately from 8.4°N to 23.3°N and from 21.5°E to 39.0°E, and covers an area of about $1.88 \times 10^6$ km$^2$ (Arliah et al., 2021). The country spans a pronounced north–south climatic gradient.

Annual rainfall increases from less than 100 mm $yr^{-1}$ across the desert north to about 700–1,200 mm $yr^{-1}$ in the wetter south (Arliah et al., 2021; Red Cross Red Crescent Climate Centre, 2024). Vegetation broadly follows this climatic gradient, ranging from sparse desert vegetation in the north to increasingly continuous grassland, savanna, and woodland vegetation toward the south.

Burned-area observations were obtained from the Moderate Resolution Imaging Spectroradiometer (MODIS) MCD64A1 Collection 6.1 product (Giglio et al., 2018). MCD64A1 is a monthly Level-3 product that records the estimated day of burn for each detected burned grid cell on a nominal 500-m MODIS sinusoidal grid. The actual grid-cell spacing is approximately 463.31 m. We analyzed data from August 2002 to July 2025 using six MODIS tiles intersecting Sudan: h20v06, h21v06, h20v07, h21v07, h20v08, and h21v08. For each month, the Burn Date layer was extracted, and the six tiles were mosaicked and clipped to the national boundary. Only positive Burn Date values ($>0$) were used as burn detections and converted from ordinal day of year to absolute calendar dates. Zero values and missing or invalid observations were excluded from event connectivity. Excluding missing or invalid observations from event construction does not establish that the corresponding locations were unburned. The monthly rasters were then concatenated along a continuous daily time axis. This procedure allowed burned grid cells to be linked across tile boundaries, month boundaries, and calendar years during fire-event delineation.

Near-surface wind fields were obtained from the European Centre for Medium-Range Weather Forecasts ERA5 atmospheric reanalysis (Hersbach et al., 2020). We retrieved hourly zonal and meridional wind components at 10 m above the surface (u10 and v10, respectively) from August 2002 to July 2025 at a horizontal resolution of 0.25°. Daily mean u10 and v10 fields were calculated from hourly data and spatially subset to Sudan. Long-term seasonal mean wind fields were then calculated by averaging the daily u10 and v10 components separately over the dry season (October–May) and wet season (June–September). These resulting mean wind-vector fields were overlaid on the spatial distributions of the fire-propagation types to provide qualitative regional atmospheric context. They were not used to quantify event-specific relationships between wind conditions and fire propagation.

## 2.2 Identification of fire events

Fire-event delineation was implemented using FIREDpy version 2.0.0 (Balch et al., 2020). For the selected sp3–tp3 configuration, the spatial neighborhood extended up to three MODIS grid cells from the focal cell in each of the $x$ and $y$ directions, forming a square neighborhood. At a grid-cell spacing of approximately 463.31 m, this corresponds to approximately 1.39 km along either grid axis. Burned cells within this neighborhood were temporally connected when the absolute difference between their calendar burn dates was $\leq$3 days.

Each spatiotemporal coordinate was tracked to prevent duplicate assignment. When a newly evaluated neighborhood connected multiple existing events, their records were merged, the associated coordinates were assigned to the retained event identifier, and redundant event records were removed. Event labels were propagated through overlapping neighborhoods so that directly or transitively connected burned cells belonged to the same event. These parameters therefore defined local connectivity rather than the maximum size or duration of an event. Processing the complete record as a continuous space-time dataset prevented events from being artificially divided at MODIS tile boundaries, month boundaries, or calendar-year boundaries.

The sensitivity to the spatial and temporal connectivity parameters was examined using Global Fire Atlas event ID 696293, which burned from 27 October to 27 November 2004, as a reference case (Andela et al., 2019). Five parameter combinations were tested: sp1–tp1, sp1–tp3, sp3–tp1, sp3–tp3, and sp3–tp7, where sp denotes the maximum spatial separation in MODIS grid cells and tp denotes the maximum temporal separation in days. For each parameter combination, the number, spatial extent, and configuration of the delineated events were visually compared with the Global Fire Atlas event footprint. A parameter combination was considered overly restrictive when it divided the reference event into several smaller events and overly permissive when it connected the reference event to surrounding burned areas outside its footprint. Two additional events from different locations and years, with Global Fire Atlas burned areas of 122.19 and 725.64 $km^2$, were examined under five spatial–temporal connectivity configurations (Supplementary Figure S1). These comparisons illustrate the sensitivity of event fragmentation and delineated extent to connectivity settings. Because both approaches use MCD64A1 burned-area observations, the assessment provides a comparison of event-delineation consistency rather than independent observational validation. Its implications are restricted to the cases and conditions examined.

The sp3–tp3 combination was selected as a practical compromise between event fragmentation and excessive merging. The spatial parameter corresponds to three MODIS grid cells, or approximately 1.39 km, and the temporal parameter corresponds to 3 days. This parameter combination was applied consistently to the complete Sudan record from August 2002 to July 2025. To restrict subsequent analyses to events with sufficient spatial and temporal development for estimating mean areal spread rate and displacement, events with durations of less than 3 days or burned areas of less than 20 $km^2$ were excluded. After filtering, 13,290 fire events were retained for the subsequent spatial, propagation-type, and temporal trend analyses.

## 2.3 Fire-event metrics and statistical analyses

For each fire event $i$, the ignition date ($t_{i,\mathrm{ign}}$) and final burn date ($t_{i,\mathrm{end}}$) were defined as the earliest and latest burn dates, respectively, among all burned grid cells assigned to the event. Event duration was calculated inclusively using Equation 1 as follows:

$$T_i = t_{i,\mathrm{end}} - t_{i,\mathrm{ign}} + 1 \quad (1)$$

where $T_i$ is expressed in days. Total event area ($A_i$, $km^2$) was calculated as the sum of the areas of all burned MODIS grid cells

assigned to that event. The mean areal spread rate ($R_i$) was then calculated using Equation 2 as follows:

$$R_i = \frac{A_i}{T_i} \quad (2)$$

where mean areal spread rate is expressed in km$^2$ day$^{-1}$. This metric represents the mean rate of burned-area accumulation over the event duration.

To characterize event displacement, burned grid cells were grouped by event identification number and burn date to construct daily burned-area polygons for each event. The ordered sequence of unique burn dates for each event was divided into early and late halves. Polygons within each half were merged, and the centroid of the resulting burned area was calculated separately for the early and late portions of the event. Centroids of the merged burned-area polygons representing the early and late halves of the ordered sequence of unique burn dates were calculated in the MODIS sinusoidal projected coordinate system. Signed zonal and meridional displacements were calculated using Equation 3 as follows:

$$\Delta x_i = x_{i,late} - x_{i,early}, \quad \Delta y_i = y_{i,late} - y_{i,early} \quad (3)$$

where positive $\Delta x_i$ and $\Delta y_i$ indicate eastward and northward displacement, respectively. Net centroid displacement was calculated using Equation 4 as follows:

$$D_i = \sqrt{\Delta x_i^2 + \Delta y_i^2} \quad (4)$$

The displacement direction was derived from the azimuth of the same vector. Mean areal spread rate is defined as the total burned area of a fire event divided by its duration (km$^2$ d$^{-1}$). It describes event-averaged burned-area accumulation and does not measure the linear velocity of an advancing fire front. "Expansion" refers to the increase in burned area as an event develops. "Centroid displacement" denotes the net spatial shift between the centroids calculated for the early and late periods of an event; it does not represent the distance traveled by the fire front. "Movement" describes this centroid-based shift and its direction, whereas "propagation" is used as a broader descriptive term for event-level spatial and temporal development, encompassing areal expansion and centroid displacement. Only events with valid early- and late-period centroids were included in the displacement analysis Events with $D_i < 2$ km were excluded from the propagation-type analysis to reduce instability associated with small centroid displacements.

For spatial distribution, each event was assigned to a 0.25° grid cell according to its event centroid. Fire-event density in grid cell j was calculated using Equation 5 as follows:

$$\rho_j = \frac{N_j}{A_j \times 23} \times 1,000 \quad (5)$$

where $N_j$ is the number of events assigned to the cell and $A_j$ is its area in km$^2$. Event density was therefore expressed as events per 1,000 km$^2$ yr$^{-1}$. Median event area and median areal spread rate were also calculated for each grid cell containing valid events. For events with valid displacement estimates, displacement directions were summarized using polar histograms with 16 directional sectors of 22.5° each. Events in each sector were further grouped into areal spread rate classes of <0.25, 0.25–0.5, 0.5–1, 1–2, 2–5, 5–10, and ≥10 km$^2$ d$^{-1}$.

Broad fire-propagation types were identified using $K$-means clustering of events with valid movement estimates ($D_i \geq 2$ km). The input variables were signed zonal displacement, signed meridional displacement, and transformed mean areal spread rate, $L_i = \ln(1 + R_i)$, with $R_i$ expressed in km$^2$ d$^{-1}$. The two displacement components were divided by a common scale,

$$s_D = \sqrt{\frac{Var(\Delta x) + Var(\Delta y)}{2}} \quad (6)$$

without centering. The transformed spread-rate variable was centered and scaled using its mean, ($\bar{L}$), and standard deviation, ($s_L$). The resulting feature vector was

$$X_i = \left( \frac{\Delta x_i}{s_D}, \frac{\Delta y_i}{s_D}, \lambda \frac{L_i - \bar{L}}{s_L} \right), \quad (7)$$

where ($\lambda = 1$) for the primary analysis. Scaling statistics were calculated across the events retained for clustering, using population variances and standard deviations with divisor $N$, where $N$ is the number of events included in clustering.

$K$-means clustering was implemented using scikit-learn version 1.7.2, with 40 random initializations and a fixed random seed of 20260729 (Pedregosa et al., 2011). Candidate solutions with $K = 2$–8 were evaluated. The selected solution maximized the silhouette coefficient for ($\lambda = 1$), with silhouette scores calculated from a random sample of 4,000 events using the same seed. Within-cluster sums of squared errors were also calculated as complementary diagnostics. Given this limited separation, the clusters are interpreted as broad descriptive groupings based on mean areal spread rate and centroid displacement, rather than as distinct fire regimes. The resulting clusters were designated as the lower-expansion and higher-expansion types according to their relative areal spread rate and movement characteristics. The two types were then summarized in terms of displacement, areal spread rate, burned area, duration, and displacement direction.

For each candidate $K$, classification stability was assessed using 20 resampling replicates. Each replicate contained 8,000 events sampled with replacement from the clustering population. The feature transformation and scaling were held fixed at the values calculated from the full clustering population. $K$-means was fitted to each replicate using 10 initializations, and the fitted model was used to assign labels to all retained clustering events. Agreement with the corresponding full-population solution was quantified using the adjusted Rand index (ARI), summarized by its mean and minimum across replicates. Sensitivity to feature weighting was also assessed by comparing the classifications obtained with ($\lambda = 0.5$) and ($\lambda = 2$) against the primary solution with ($\lambda = 1$), using ARI at the same $K$.

Temporal analyses were conducted over 23 fire seasons, each extending from 1 August to 31 July of the following year. Each event was assigned to an August–July fire season according to its ignition date. For each of the 23 fire seasons, we calculated the 10th percentile, median, and 90th percentile of mean areal spread rate from the retained events. Each quantile therefore formed a time series of 23 seasonal summary values. Long-term trends

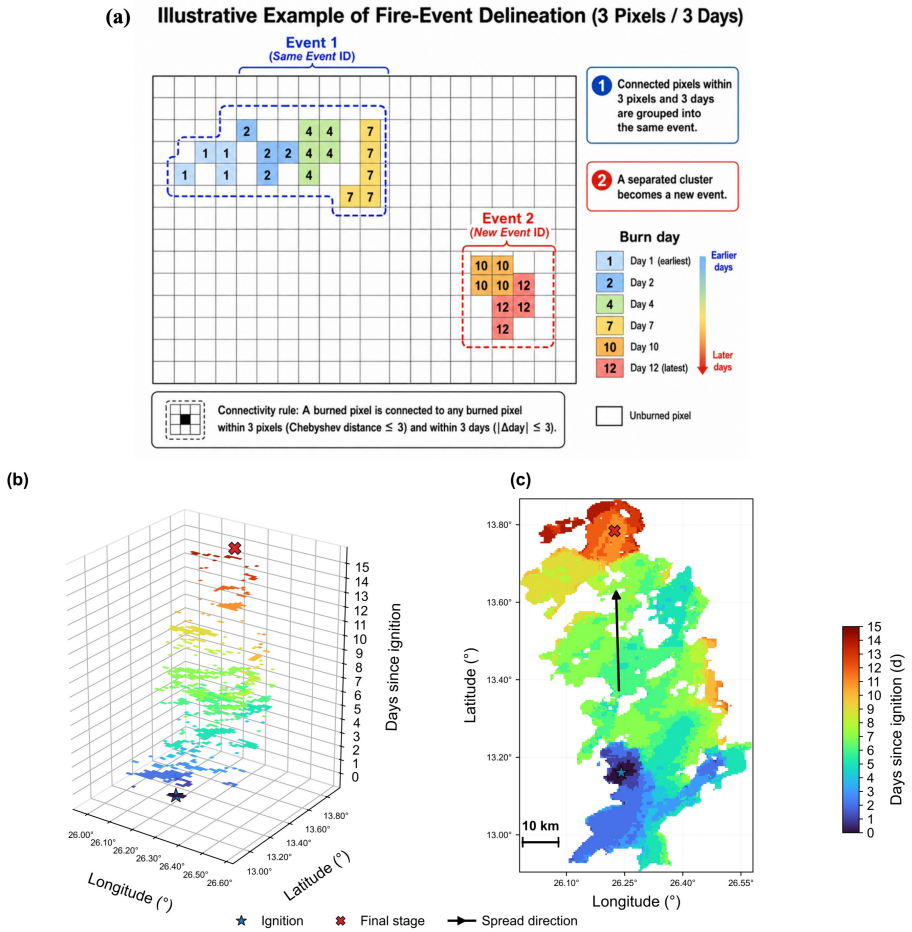



FIGURE 1
Spatiotemporal reconstruction of a selected fire event delineated from the MODIS MCD64A1 burned-area product using spatial and temporal connectivity thresholds of 3 pixels and 3 days, respectively. **(a)** Illustrative of the event delineation procedure, in which burned pixels connected in space and time are grouped into the same event, whereas a spatially disconnected cluster is assigned a different event ID. **(b)** Three-dimensional representation of event evolution in longitude-latitude-time space from 1 to 16 October 2007. **(c)** Top-view of event progression, with colors indicating days since ignition. The blue star and red cross indicate the estimated ignition location and final record locations, respectively, and the black arrow indicates the overall event-progression direction.

were estimated by applying the Theil–Sen estimator to $\log_{10}$-transformed seasonal values, and trend significance was assessed using the Mann–Kendall test. Individual events were not treated as independent temporal replicates in these trend analyses. For the median and percentile series of all events, the fitted log-scale trends were back-transformed and expressed as percentage change per decade. For comparisons between propagation types and seasons, the corresponding multiplicative trends were converted to absolute changes relative to the median seasonal areal spread rate and reported in $km^2\ day^{-1}\ decade^{-1}$. Areal spread rate distributions were additionally compared between the early period (2002/03–2013/14) and the late period (2014/15–2024/25). Distributional differences were visualized using kernel-density estimates and evaluated using the two-sample Kolmogorov–Smirnov test.

Each fire event was classified seasonally according to the number of calendar days in its inclusive duration. Days from October through May were assigned to the dry season, whereas days from June through September were assigned to the wet season. An event was classified as dry-season or wet-season when most of its duration occurred within the corresponding season. Events containing equal numbers of dry- and wet-season days were classified as transitional. Transitional events were retained in analyses of all events but excluded from direct comparisons between the dry and wet seasons. Fire-event density was then mapped by propagation type and seasonal class and compared qualitatively with the corresponding long-term mean 10-m wind fields. The analyses were conducted in Python 3.10.19 using NumPy 2.2.6, pandas 2.3.3, SciPy 1.15.2, and scikit-learn 1.7.2.

# 3 Results

## 3.1 Fire-event delineation and climatological characteristics

The illustrative fire event spanned 16 days, from 1 to 16 October 2007. Burned area was first detected in the southern portion of the event and subsequently progressed northward (Figure 1). A separate event from the Global Fire Atlas was used to examine

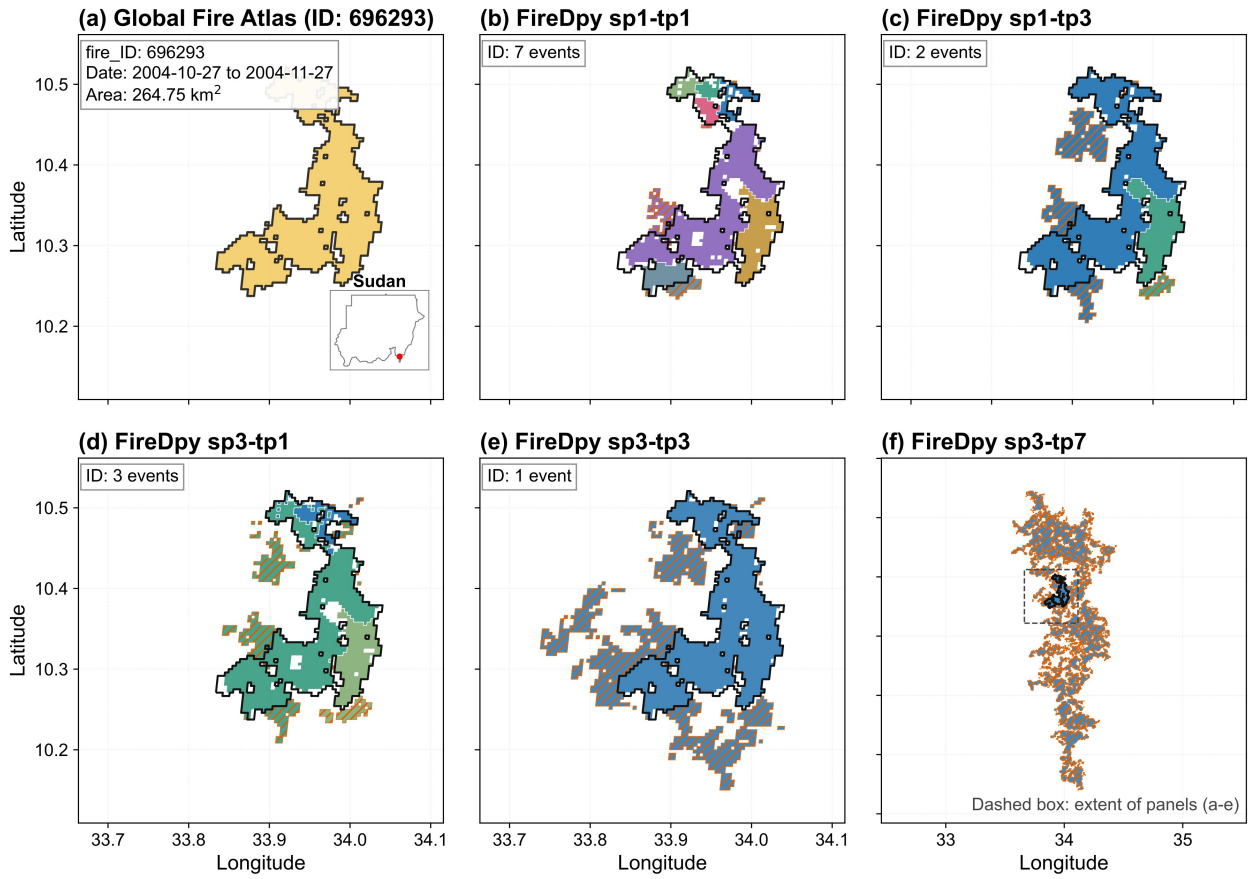



FIGURE 2
Sensitivity of fire-event delineation to alternative spatial and temporal connectivity thresholds, evaluated using a Global Fire Atlas event as a reference case. **(a)** Reference-event boundary from the Global Fire Atlas. **(b–e)** Event delineated using the sp1–tp1, sp1–tp3, sp3–tp1, and sp3–tp3 settings, respectively, where sp and tp denote the spatial threshold in pixels and temporal threshold in days. Colors distinguish separately delineated events, and orange hatching indicates burned pixels assigned to events beyond the reference-event boundary. **(f)** Result obtained using the sp3–tp7 setting, shown over a wider spatial extent; the dashed box indicates the extent shown in panels **(a–e)**.

the sensitivity of event delineation to the spatial and temporal connectivity parameters (Figure 2). The sp1–tp1, sp1–tp3, and sp3–tp1 parameter combinations divided the reference event into seven, two, and three events, respectively. In contrast, the sp3–tp3 setting detected the reference footprint as a single coherent event, whereas sp3–tp7 incorporated adjacent burned areas beyond the reference footprint. The sp3–tp3 combination was therefore selected as the practical compromise between event fragmentation and excessive merging and was applied to the complete Sudan record.

Applying the sp3–tp3 parameter combination and the duration and area filters yielded 13,290 fire events. Fire-event density was highest in southern Sudan, particularly between approximately 9°N and 14°N, and declined markedly toward the north. Few events were identified north of approximately 16°N (Figure 3a). Event area followed a strongly right-skewed distribution, with small retained events occurring much more frequently than large events (Figure 3b). The median event area was 37.57 km$^2$, whereas the 95th percentile was 280.03 km$^2$.

Across the 23 fire seasons, total burned area varied substantially but declined significantly at a rate of 20,721.31 km$^2$ decade$^{-1}$ ($p < 0.001$; Figure 3c). By contrast, the annual mean event area showed no significant trend, with a Theil-Sen slope of −2.31 km$^2$ decade$^{-1}$ ($p = 0.734$; Figure 3d). Thus, the long-term decline in total burned area was not accompanied by a detectable change in mean event area.

Event-level areal spread rate also followed a strongly right-skewed distribution (Figure 4a). The median areal spread rate was 4.42 km$^2$ d$^{-1}$, and the 95th percentile was 18.39 km$^2$ d$^{-1}$. Of the 13,290 events, 12,438 yielded valid centroid-displacement estimates and were included in the directional analysis. Centroid-displacement directions occurred across all 16 sectors, with relatively higher frequencies toward the north and east (Figure 4b). Spatially, grid-cell medians of event-level mean areal spread rates were highest mainly in the western and central portions of the active fire belt, with additional high-rate cells scattered farther east (Figure 4c).

## 3.2 Cluster-derived fire-propagation types

Of the 13,290 filtered fire events, 9,274 (69.8%) yielded valid centroid-displacement estimates and had a net movement distance of at least 2 km. These events were included in the $K$-means cluster analysis. The within-cluster sum of squared errors decreased monotonically as $K$ increased from 2 to 8, whereas the mean silhouette score was highest for $K = 2$

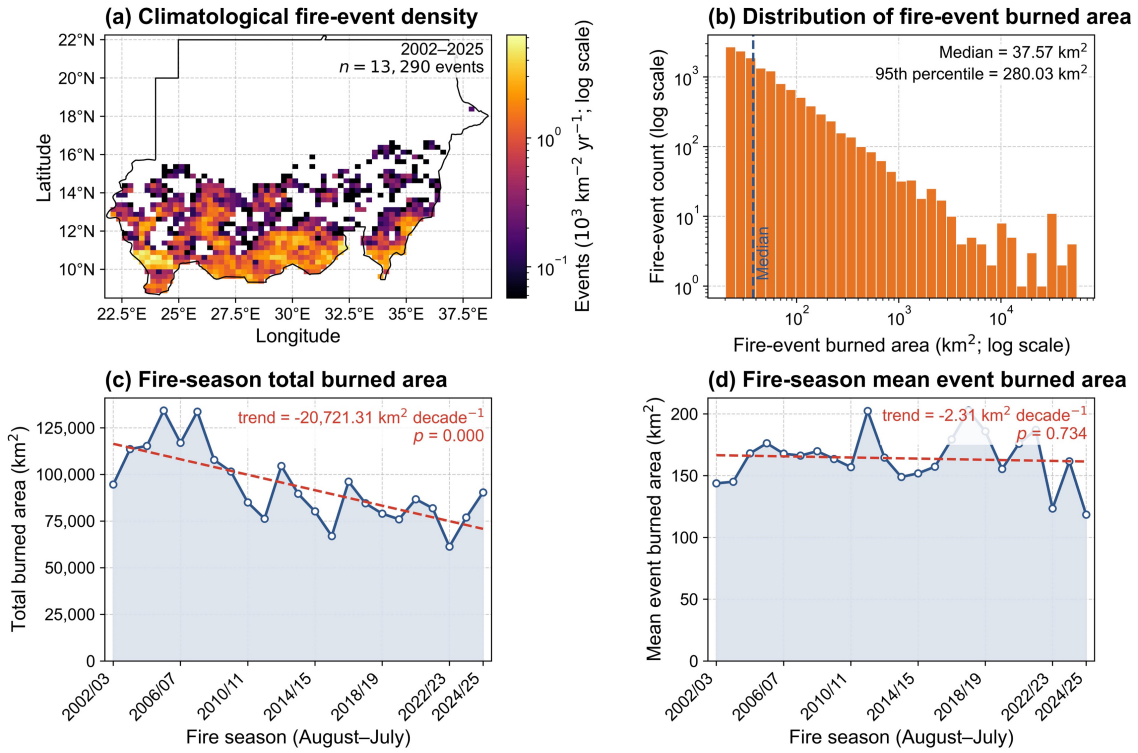



FIGURE 3
Spatial distribution, event-size distribution, and interannual burned-area dynamics of 13,290 fire events across Sudan during the 2002/03–2024/25 fire seasons. Events were retained when duration was ≥3 days and total burned area was ≥20 km$^2$. **(a)** Climatological fire-event density calculated on a 0.25° grid and expressed as events per 1,000 km$^2$ yr$^{-1}$ using a logarithmic color scale. Event density is displayed on a logarithmic color scale to show spatial variation across its wide range of values. **(b)** Distribution of fire-event burned area; the dashed line marks the median (37.57 km$^2$), and the 95th percentile is 280.03 km$^2$. **(c)** The time series of total burned area. **(d)** The time series of mean burned area per event. Solid lines show annual fire-season values, and dashed lines indicate long-term trends. Fire seasons extend from August of 1 year to July of the following year.

(0.325; Figure 5a). Although this coefficient indicated only modest cluster separation, the two-cluster solution provided the best-supported and most parsimonious solution among the candidate values of *K*. The resulting clusters were therefore interpreted as broad propagation types rather than sharply separated fire classes.

The lower-expansion type comprised 6,450 events (69.6% of classified events), whereas the higher-expansion type comprised 2,824 events (30.4%; Figures 5b, c). Lower-expansion events had a median movement distance of 3.6 km and a median areal spread rate of 3.4 km$^2$ d$^{-1}$. By comparison, higher-expansion events had a median movement distance of 5.0 km and a median areal spread rate of 10.1 km$^2$ d$^{-1}$. Thus, the higher-expansion type had a nearly threefold higher median areal spread rate and a 39% greater median displacement distance. The spatial distributions of the two types broadly overlapped, with both concentrated mainly in southern Sudan.

The two propagation types also differed in their movement directions and areal spread rate distributions (Figure 6). Lower-expansion events occurred most frequently in the northern and eastern sectors and intermediate areal spread rate classes accounted for the largest shares of these events. Higher-expansion events showed a broader directional distribution, with relatively high frequencies across the eastern, southeastern, southern, and western sectors. The ≥10 km$^2$ d$^{-1}$ class accounted for the largest share of higher-expansion events.

Differences were also apparent in other event-level attributes (Figure 7). Consistent with the variables used for clustering, higher-expansion events had greater centroid displacement and areal spread rate than lower-expansion events. They also had larger event areas but slightly shorter durations. Nevertheless, the interquartile ranges overlapped substantially for several attributes, consistent with the modest silhouette coefficient and indicating that the two types represent broad tendencies rather than discrete fire regimes.

## 3.3 Temporal changes in fire-event spread dynamics

Long-term trends in event-level areal spread rate varied across the spread-rate distribution over the 23 fire seasons (Figures 8a–c). The median areal spread rate decreased by 9.43% decade$^{-1}$ ($p < 0.001$; Figure 8a), while the 10th percentile of areal spread rate declined by 9.44% decade$^{-1}$ ($p < 0.001$; Figure 8c). The 90th percentile of areal spread rate showed a weaker decline of 5.34% decade$^{-1}$, but the trend was not statistically significant ($p = 0.091$; Figure 8b). Thus, statistically detectable declines occurred in the lower and central portions of the spread-rate distribution, but not in its upper tail.

Consistent with these temporal trends, the areal spread rate distribution shifted toward lower values between the early and late

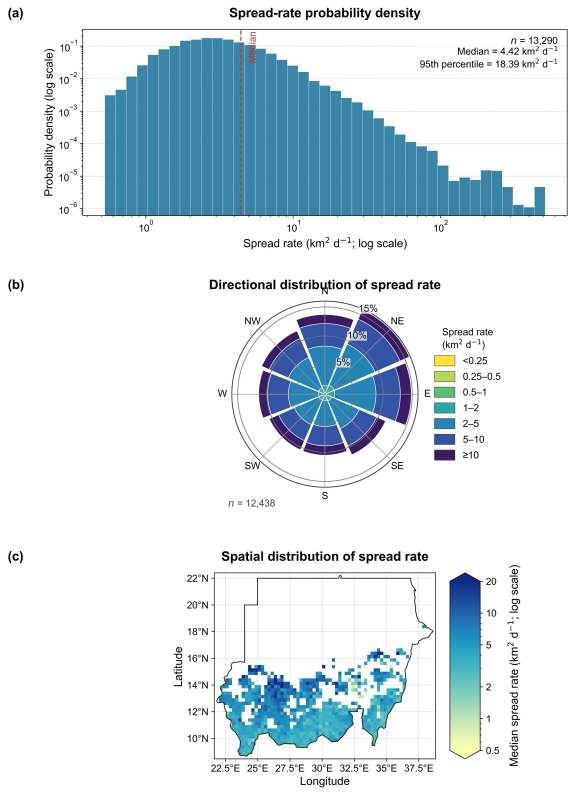



FIGURE 4
Areal spread rates and movement characteristics of fire events across Sudan during 2002–2025. **(a)** Probability-density distribution of areal spread rate for 13,290 filtered fire events; the dashed line marks the median (4.42 km$^2$ d$^{-1}$), and the 95th percentile is 18.39 km$^2$ d$^{-1}$. **(b)** Directional distribution of event movement for 12,438 events with valid movement estimates. Radial frequencies represent event counts within directional sectors, and stacked colors denote areal spread rate classes of <0.25, 0.25–0.5, 0.5–1, 1–2, 2–5, 5–10, and ≥10 km$^2$ d$^{-1}$. **(c)** Spatial distribution of median event areal spread rate calculated within 0.25° grid cells and displayed using a logarithmic color scale.

study periods (Figure 8d). The median areal spread rate during 2014/15–2024/25 was 12.0% lower than that during 2002/03–2013/14, and the two distributions differed significantly according to the two-sample Kolmogorov–Smirnov test ($p < 0.001$).

Long-term trends also differed between the two propagation types (Figures 9a, c). The median areal spread rate of higher-expansion events showed no statistically detectable trend ($-0.04$ km$^2$ d$^{-1}$ decade$^{-1}$, $p = 0.958$; Figure 9a). By contrast, the corresponding rate for lower-expansion events declined significantly by $-0.24$ km$^2$ d$^{-1}$ decade$^{-1}$ ($p < 0.001$; Figure 9c).

Further stratification by season showed that the significant decline was confined to dry-season lower-expansion events (Figures 9b, d). Higher-expansion events showed no statistically significant trend in either the dry season ($-0.04$ km$^2$ d$^{-1}$ decade$^{-1}$, $p = 0.917$) or wet season ($+2.30$ km$^2$ d$^{-1}$ decade$^{-1}$, $p = 0.260$). For Lower-expansion events, areal spread rate declined significantly during the dry season ($-0.24$ km$^2$ d$^{-1}$ decade$^{-1}$, $p < 0.001$), whereas no significant trend was detected during the wet-season ($+0.73$ km$^2$ d$^{-1}$ decade$^{-1}$, $p = 0.112$). The wet-season estimates were based on substantially fewer events and should therefore be interpreted cautiously.

Events of both propagation types occurred overwhelmingly during dry season (Figure 10). Among higher-expansion events, 2,779 occurred overwhelmingly during the dry season and 40 during the wet season. The corresponding numbers for lower-expansion events were 6,380 and 64, respectively. During the dry season, both types were concentrated mainly across southern Sudan, although lower-expansion events formed a broader and denser spatial belt. Wet-season events were sparse and spatially discontinuous. The accompanying long-term mean 10-m wind fields provide regional atmospheric context and do not represent the wind conditions experienced by individual fire events.

# 4 Discussion

## 4.1 Uneven changes in fire-event dynamics

By resolving burned area into discrete fire events, this study shows that long-term changes in Sudanese fire activity were uneven across the spread-rate distribution, propagation types, and seasons. Total burned area declined without a detectable change in mean event area, while event-level mean areal spread rate decreased primarily in the lower and central portions of its distribution. This decline was concentrated in lower-expansion events during the dry season, whereas no statistically significant trend was detected for higher-expansion events. These results indicate that the long-term change was characterized by a contraction in the spread of more frequent, lower-expansion events rather than a uniform weakening of fire propagation. Such differentiation cannot be resolved from aggregate burned-area statistics alone (Andela et al., 2019; Mahood et al., 2025).

The contrasting trends in total burned area and mean event area indicate that aggregate and event-level changes can become decoupled. Declining burned area has been widely reported across African savannas and has been associated with changes in rainfall, agricultural expansion, landscape fragmentation, fire-season length, and land management (Andela and van der Werf, 2014; Andela et al., 2017; Forkel et al., 2019; Kelley et al., 2019; Yin et al., 2020; Jones et al., 2022; Mohammed et al., 2026). These processes may also be relevant to Sudan, but they were not evaluated in the present study. The interpretation supported here is therefore narrower: the decline in total burned area was not accompanied by a systematic reduction in the mean area of the fire events. Changes in the average event size alone cannot account for the observed decline in aggregate burned area.

Changes within the distribution of mean areal spread rate were similarly uneven. The comparable declines in the median and 10$^{th}$ percentile, both approximately 9.4% decade$^{-1}$, indicate that the shift toward lower values extended across typical and relatively slow-expanding events. The 12.0% reduction in median spread rate between the early and late study periods provides further evidence of this distributional shift. By contrast, the weaker and non-significant trend at the 90$^{th}$ percentile leaves the long-term behavior of rapidly expanding events unresolved. It should not be interpreted as evidence that upper-tail events remained unchanged. Nor does

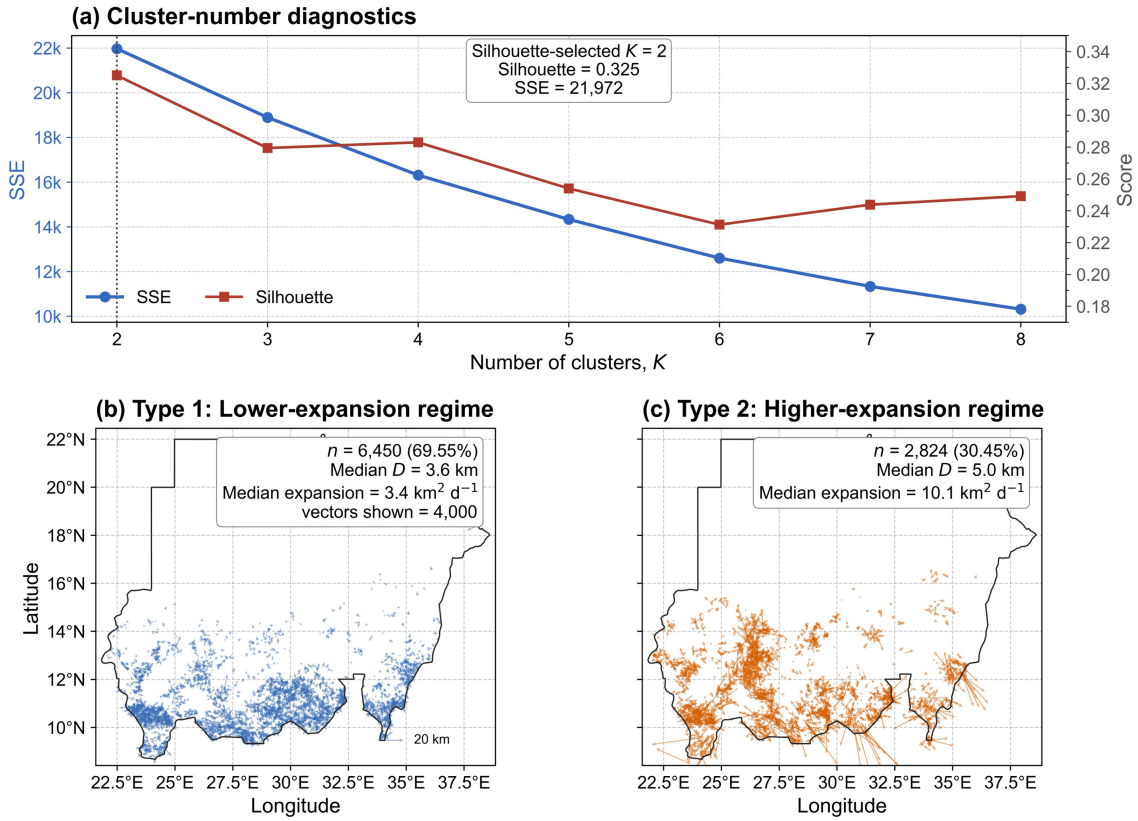



FIGURE 5
Fire-propagation types identified by $K$-means clustering for events with valid movement estimates and net movement distance ≥2 km. **(a)** Cluster diagnostics for $K = 2–8$, including the within-cluster sum of squared errors and silhouette score. Clustering was based on three standardized variables: signed zonal displacement, signed meridional displacement, and $\log_{10}$-transformed event-level areal spread rate. The silhouette score was highest at $K = 2$. **(b)** Lower-expansion type, comprising 6,450 events (69.6%), with a median movement distance of 3.6 km and a median areal spread rate of 3.4 km$^2$ d$^{-1}$. **(c)** Higher-expansion type, comprising 2,824 events (30.4%), with a median movement distance of 5.0 km and a median areal spread rate of 10.1 km$^2$ d$^{-1}$. Arrows represent displacement from the early-half centroid to the late-half centroid of each event; up to 4,000 vectors are displayed per type for visual clarity. Only events with duration ≥3 days, burned area ≥20 km$^2$, and $D \geq 2$ km were included in the clustering analysis.

the decline in the lower part of the distribution necessarily indicate a reduction in meteorological fire danger. Event expansion reflects the combined influence of weather, fuels, landscape structure, and human activity, whereas fire-weather indices describe atmospheric potential rather than the realized development of individual fires (Juang et al., 2022; Cunningham et al., 2024; Jones et al., 2022; Jolly et al., 2015; Abatzoglou et al., 2019).

## 4.2 Propagation types and seasonal-spatial patterns

Stratifying events by their propagation characteristics further localized the observed decline. Higher-expansion events combined faster areal spread with greater centroid displacement and larger burned areas, whereas lower-expansion events were more frequent and occupied a broader spatial belt. The modest silhouette score indicates, however, that these types are not sharply separated natural classes. They are better interpreted as an operational partition of a continuous range of fire-event behavior. Within this partition, the long-term decline was concentrated in lower-expansion events during the dry season. Seasonal changes in fuel moisture, fuel connectivity, vegetation flammability, and landscape fragmentation provide plausible explanations for this pattern (Moustakas and Davlias, 2021; Takacs et al., 2021; Eames et al., 2023; Mohammed et al., 2026), but distinguishing among these influences lies beyond the scope of the present analysis. Wet-season trends also require caution because they were based on relatively few events and were not statistically significant.

The geographical patterns provide additional context for the temporal changes. Fire events were concentrated across southern Sudan, where greater rainfall supports more continuous grassland, savanna, and woodland fuels than in the arid north. This distribution agrees with regional observations and with the established roles of fuel availability and seasonal moisture in savanna fire activity (Archibald et al., 2009; Lehmann et al., 2014; Modawi et al., 2015; Mohieldeen and Hassan, 2025). Higher median areal spread rates in parts of western and central Sudan may similarly reflect regional differences in fuel continuity, moisture conditions, land cover, or human land use, although these relationships were not tested. Net movement occurred across all directional sectors, with only modest concentrations toward the north and east, suggesting that fire events did not exhibit a single

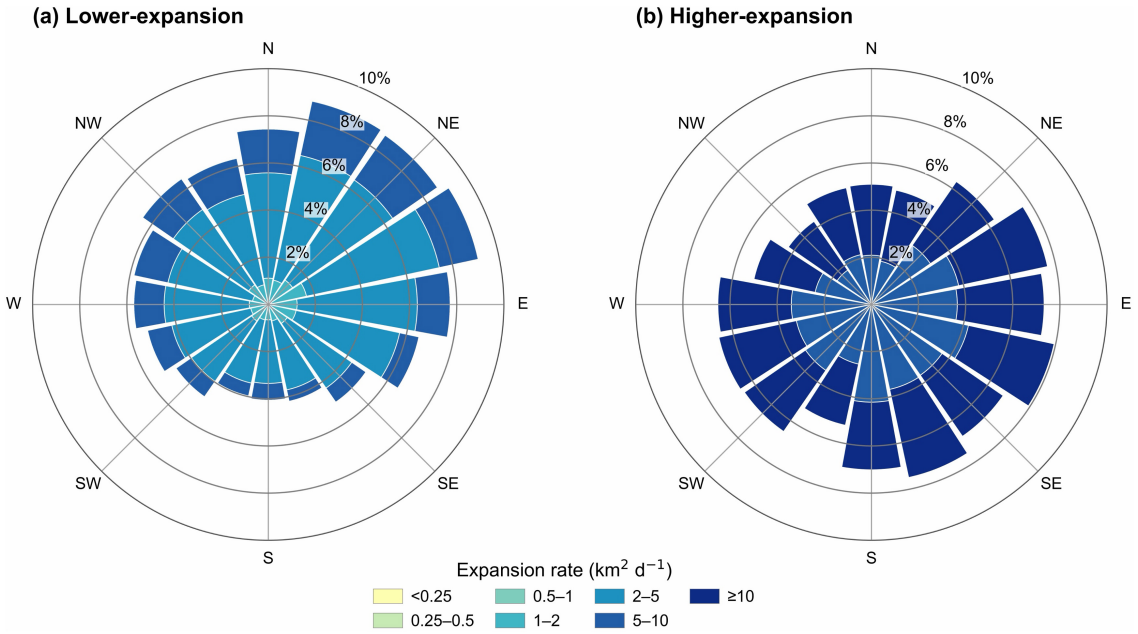



FIGURE 6
Directional distributions of fire-event movement for the **(a)** lower-expansion and **(b)** higher-expansion propagation types across Sudan during 2002–2025. Polar histograms show the percentage of events within 16 directional sectors. Stacked colors denote areal spread rate classes of <0.25, 0.25–0.5, 0.5–1, 1–2, 2–5, 5–10, and ≥10 km$^2$ d$^{-1}$. Both panels use the same radial percentage scale to facilitate direct comparison between the two propagation types.

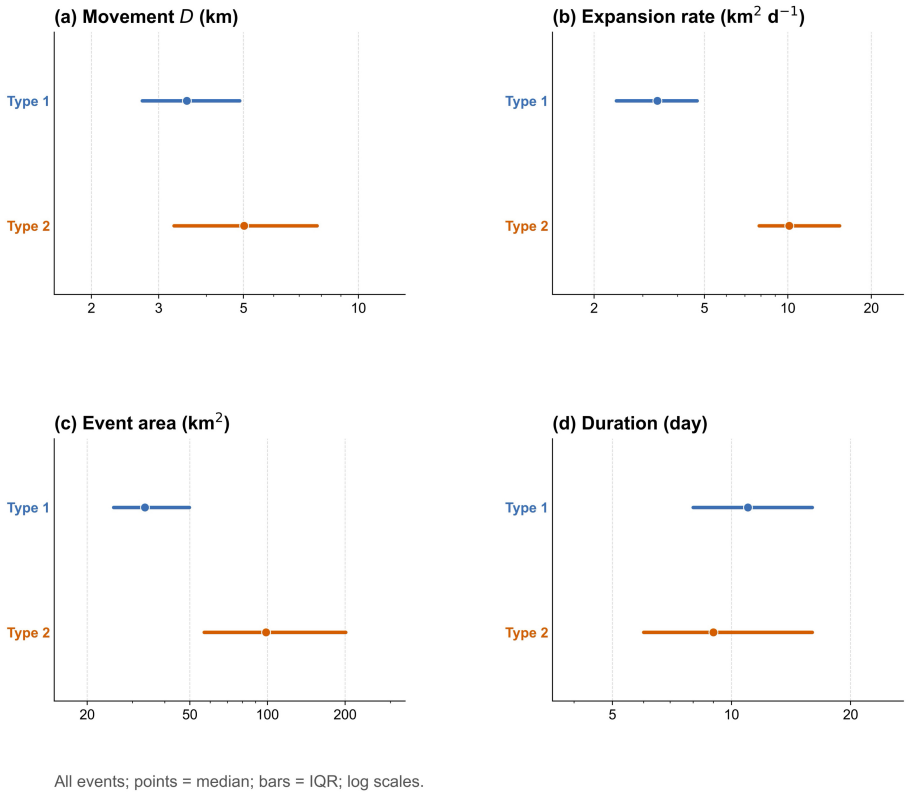



FIGURE 7
Comparison of event-level attributes between the lower-expansion and higher-expansion fire-propagation types across Sudan during 2002–2025. Panels show **(a)** net movement distance (km), **(b)** event-level areal spread rate (km$^2$ d$^{-1}$), **(c)** burned area (km$^2$), and **(d)** event duration (days). Points represent medians, and horizontal bars indicate interquartile ranges. All *x*-axes are displayed on logarithmic scales to improve the readability of their strongly right-skewed distributions.

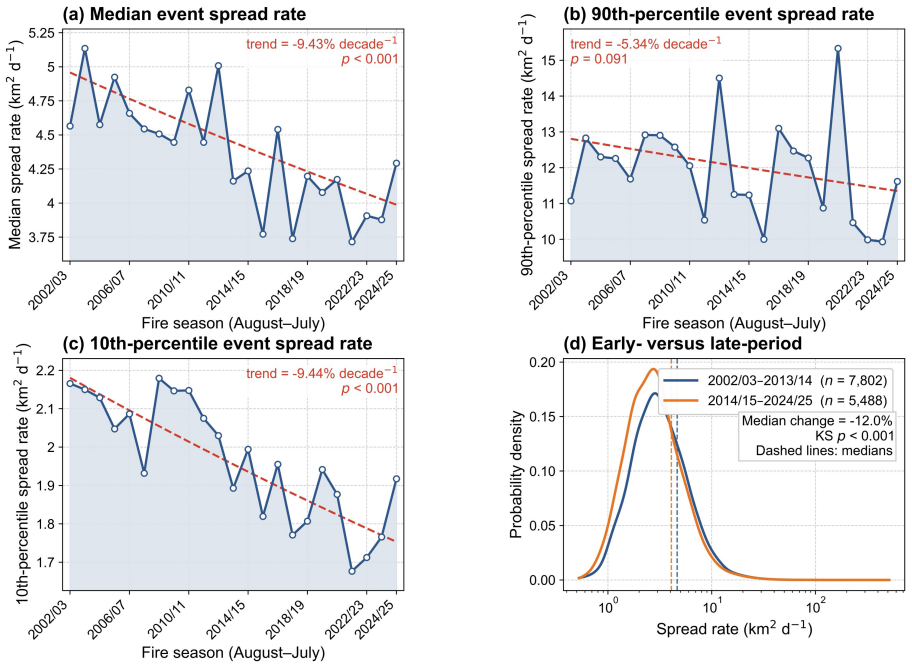



FIGURE 8
Long-term changes in event-level areal spread rates across Sudan during the 2002/03–2024/25 fire seasons. Panels show annual values of the **(a)** median, **(b)** 90th-percentile, and **(c)** 10th-percentile. Blue lines and circles represent fire-season values, and red dashed lines show Theil–Sen trends; annotations report relative change per decade and the corresponding trend-test $p$-values. **(d)** Probability-density distributions of event areal spread rates during the early period (2002/03–2013/14; $n$ = 7,802) and late period (2014/15–2024/25; $n$ = 5,488). Vertical dashed lines indicate the corresponding period medians. Fire seasons extend from August of 1 year to July of the following year. Panel **(d)** is displayed on a logarithmic $x$-axis.

dominant direction of displacement at the national scale. The ERA5 seasonal wind fields should therefore be interpreted only as regional atmospheric context, not as evidence that the movement of individual events was controlled by the mapped winds.

## 4.3 Methodological constraints and implications

The interpretation of these patterns depends partly on how individual events were reconstructed from the MODIS burn-date record. The selected 16-day event (Figure 1b) illustrates that spatiotemporal connectivity can preserve a coherent sequence of fire development, while the comparison with the Global Fire Atlas reference event shows how restrictive thresholds fragment burned areas and wider temporal windows merge otherwise separate patches. The 3-pixel/3-day setting should consequently be regarded as a practical configuration for this regional analysis rather than a universally optimal definition of a fire (Andela et al., 2019; Balch et al., 2020). Uncertainty in MCD64A1 burn dates and its moderate spatial resolution may be particularly important for small or short-duration events (Giglio et al., 2018; Boschetti et al., 2019). The minimum burned-area and duration thresholds restrict the analysis to events covering at least 20 km$^2$ and lasting at least 3 days. These criteria preferentially retain relatively large and persistent events, excluding smaller fires that constitute an important component of African fire activity (Ramo et al., 2021). The excluded events may exhibit different temporal patterns. The choice of selection thresholds may influence the composition of the retained catalog and the resulting trend estimates. Accordingly, the findings should be interpreted as conditional on the adopted event-selection criteria and should not be generalized to all fire activity in Sudan. In addition, mean areal spread rate describes burned area accumulated per day rather than linear fire-front velocity, while centroid displacement represents net movement rather than the complete pathway of fire propagation.

The long-term trends in the 10th percentile, median, and 90th percentile of mean areal spread rate were calculated from the complete retained event population and therefore do not depend on cluster membership. In contrast, comparisons between propagation types are conditional on the adopted classification. Given the limited cluster separation, these comparisons should be interpreted as descriptive contrasts within the event population, rather than as evidence of distinct fire regimes.

Several statistical considerations constrain the interpretation of these results. Seasonal aggregation does not ensure independence between successive fire seasons, and spatial-temporal dependence among events may affect the uncertainty of seasonal summaries and distributional comparisons. The use of non-parametric methods does not itself resolve these issues. In addition, the related quantile, seasonal, and distributional comparisons require caution when interpreting individual significance tests; consistency among related analyses should not be regarded as independent confirmation. Wet-season estimates warrant particular caution

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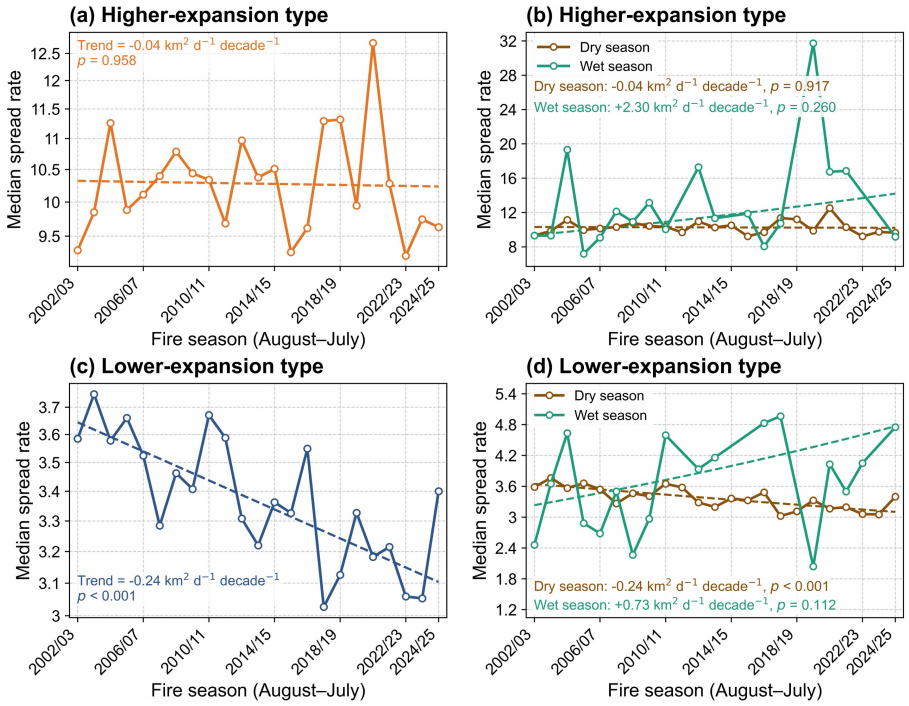



FIGURE 9
Long-term and seasonal changes in event-level areal spread rate for the higher- and lower-expansion propagation types across Sudan during the 2002/03–2024/25 fire seasons. Panels **(a, b)** show results for the higher-expansion type, and panels **(c, d)** show results for the lower-expansion type. Panels **(a, c)** present annual median areal spread rates for all events within each type, whereas panels **(b, d)** present the corresponding medians separately for dry- and wet-season events. Points and connecting lines represent fire-season medians, and dashed lines show Theil–Sen trends. Annotations report the estimated absolute change in areal spread rate (km$^2$ d$^{-1}$ decade$^{-1}$) and the corresponding trend-test $p$-values. Fire seasons extend from August of 1 year to July of the following year; the dry season spans October–May and the wet season June–September. Panels **(a, c)** are displayed on logarithmic $y$-axes.

because the smaller event samples limit precision and trend detection. Non-significant results therefore indicate insufficient evidence of a detectable trend, rather than an absence of change. The reported patterns should be interpreted with these sources of uncertainty in mind.

Taken together, the results reveal a selective rather than uniform change in Sudan's fire-event dynamics. The decline in burned area was accompanied by slower areal expansion among common lower-expansion dry-season events, while mean event area and the upper tail of the spread-rate distribution showed no statistically detectable trends. Event-based analysis therefore adds an important distributional and seasonal dimension to conventional burned-area assessments. It identifies which components of the fire-event population have changed while also showing where the available satellite record does not support a clear trend. This event-level baseline provides a basis for future work examining the climatic, ecological, and land-use processes associated with fire change across Sudan.

## 5 Conclusion

This study provides a national-scale assessment of fire-event spread dynamics across Sudan from August 2002 to July 2025 using event-based analysis of the MODIS MCD64A1 burned-area record. The identified fire events were concentrated mainly in southern Sudan and exhibited substantial variation in areal spread rate and centroid displacement. Two broad propagation types were distinguished: a more frequent lower-expansion type and a less frequent higher-expansion type characterized by faster areal spread, greater displacement, and larger burned area.

Over the 23 fire seasons, total burned area declined significantly, whereas mean event area showed no statistically detectable trend. Median and 10$^{th}$-percentile mean areal spread rates decreased by approximately 9.4% decade$^{-1}$, while the trend at the 90$^{th}$ percentile was weaker and non-significant. The decline was concentrated in lower-expansion events during the dry season, with no corresponding long-term trend detected for higher-expansion events. These contrasting responses indicate that Sudan's declining burned area coincided with a selective reduction in the areal expansion of typical fire events rather than a uniform shift across the event population. Event-based metrics therefore reveal distributional and seasonal changes that aggregate burned-area statistics alone cannot resolve. These findings apply specifically to the relatively large and persistent fire events retained in our analysis, with burned areas $\geq$20 km$^2$ and durations $\geq$3 days. Smaller or shorter-lived events were excluded, and the reported patterns should therefore not be generalized to all fire activity in Sudan. The analysis documents changes in the characteristics of the retained events but does not establish their environmental causes. The environmental factors discussed provide possible explanations for these patterns; their respective roles require further investigation.

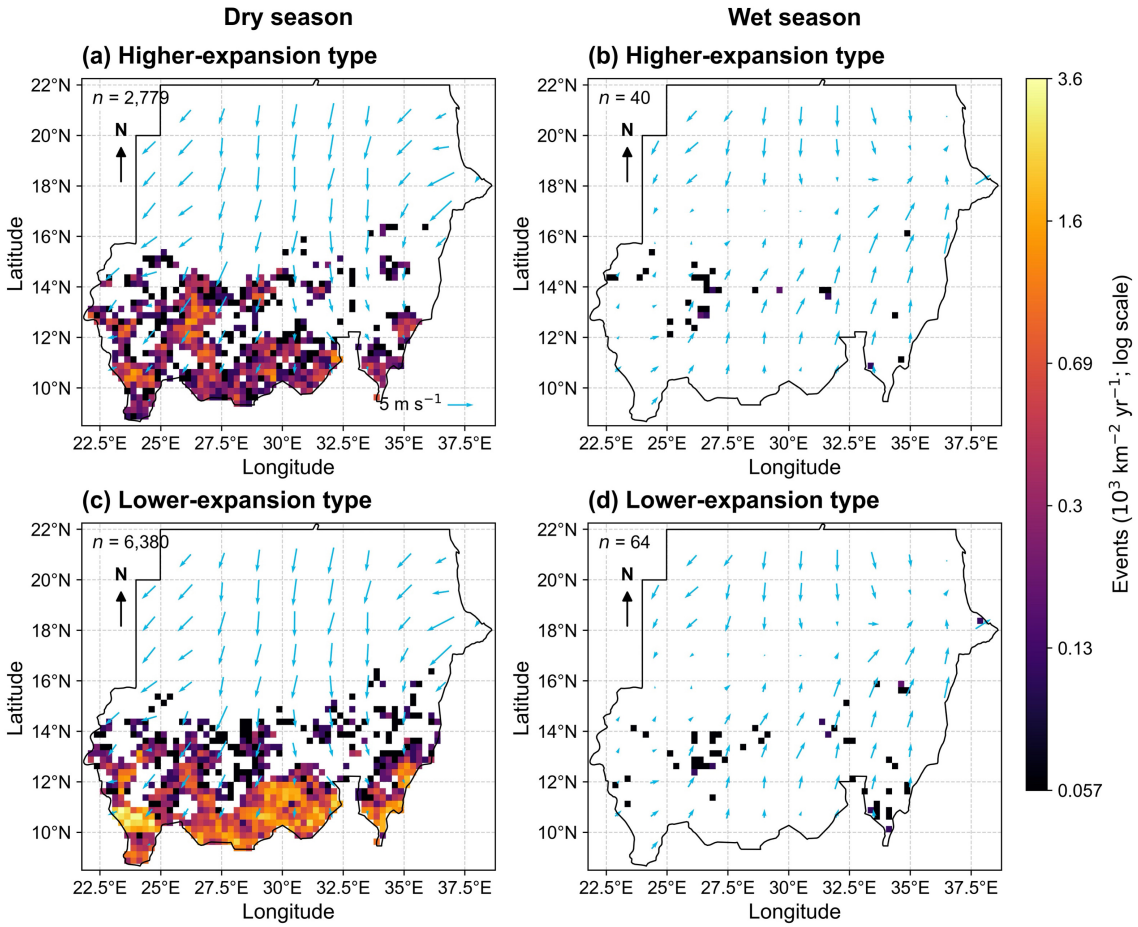



FIGURE 10
Spatial distributions of fire-event density for the higher- and lower-expansion propagation types across Sudan during 2002–2025, shown with long-term seasonal mean 10-m wind fields. Panels show **(a)** higher-expansion events during the dry season ($n$ = 2,779), **(b)** higher-expansion events during the wet season ($n$ = 40), **(c)** lower-expansion events during the dry season ($n$ = 6,380), and **(d)** lower-expansion events during the wet season ($n$ = 64). Fire-event density was calculated on a 0.25° grid and expressed as events per 1,000 km$^2$ yr$^{-1}$. Blue arrows show long-term seasonal mean 10-m wind vectors, with the reference arrow indicating 5 m/s. The dry season spans October–May and the wet season June–September. All four spatial panels use the same logarithmic color scale and color limits to facilitate direct comparison.

## Data availability statement

The study-specific scripts for FIREDpy-based event delineation and subsequent analyses are available at https://doi.org/10.5281/zenodo.22970519. The code documentation includes parameter settings, software dependencies and versions, input-data access information, and instructions for running the processing workflow.

## Author contributions

MM: Writing – original draft, Data curation, Conceptualization, Visualization, Methodology, Formal analysis. YH: Supervision, Project administration, Funding acquisition, Conceptualization, Resources, Writing – review & editing. MI: Writing – review & editing, Validation, Software.

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## Conflict of interest

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## Supplementary material

The Supplementary Material for this article can be found online at: https://www.frontiersin.org/articles/10.3389/fclim.2026.1983725/full#supplementary-material

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